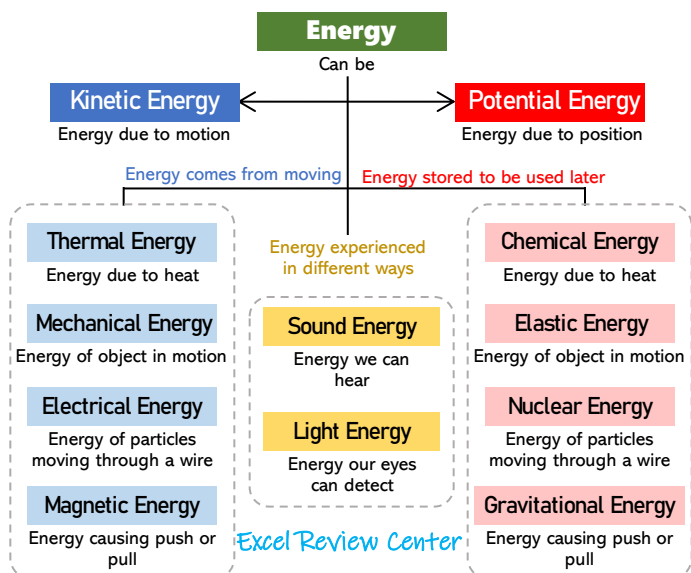




Forms of Energy



Momentum

Momentum (p) is the product of the object's mass and its velocity. It is also a vector quantity. It will always have the same direction as the direction of velocity. Normally, the term linear momentum is used to differentiate it from angular momentum. The SI unit of momentum is kg-m/s.

Momentum refers to the "amount of motion" that an object has. Since it is a function of velocity, it is a property of a moving object only. An object that is stationary has no momentum. An object that is moving, and is moving fast at that, has a large momentum. The larger the momentum an object has, the more difficult it can be stopped.

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$$\text{Momentum} = p = mv$$

Impulse

Impulse (J or Imp) refers to the change in momentum of the object. It is a function of both force and time and is also a vector quantity. The bigger the momentum change needed for an object to be stopped, it will take larger force to be applied and the time that the force is applied should also be longer. The SI unit of impulse is N-s.

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$$\text{Impulse} = F\Delta t = \Delta mv = mv_f - mv_i$$



Newton's cradle is a device that demonstrates conservation of momentum and energy using a series of swinging spheres. When one sphere at the end is lifted and released, it strikes the stationary spheres, transmitting a force through the stationary spheres that pushes the last sphere upward. The last sphere swings back and strikes the still nearly stationary spheres, repeating the effect in the opposite direction.

Important Laws

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Law of Conservation of Energy

"Energy is neither created nor destroyed, it can only be transformed from one form to another or transferred from one system to another."

Law of Conservation of Mass

"Mass is neither created nor destroyed in chemical reactions." The mass at the beginning of the reaction will equal to the mass at the end of the reaction.

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Law of Conservation of Momentum

"The total momentum before a collision is equal to the total momentum after the collision if no external forces act on the system."

Impulse – Momentum Theorem

"When a net force acts on an object, the impulse of the net force is equal to the change in momentum of the object."

Elastic & Inelastic Collisions

An **elastic collision** is a collision between two bodies in which the total kinetic energy of two bodies after the collision is equal to their total kinetic energy before the collision. Elastic collision occurs only if there is no net conversion of kinetic energy into other forms. Example: Movement of swinging balls

An **inelastic collision** is a collision in which kinetic energy is not conserved. Perfectly inelastic collision is where objects join together after the collision to form one mass. Example: Automobile collision

Elastic Collision	Inelastic Collision
Total momentum is conserved	Total momentum is conserved
Total kinetic energy is conserved	Total kinetic energy is not conserved
Forces involved are conservative forces	Forces involved are non-conservative forces
Mechanical energy is not dissipated	Mechanical energy is dissipated into heat, light, sound, etc.
Colliding objects bounce back without undergoing any deformation or heat generation	Colliding objects are distorted and heat is generated

Coefficient of Restitution

It is the negative ratio of the relative velocity between the objects after collision to the relative velocity between the objects before collision. If the coefficient of restitution is 1, then the collision is **perfectly elastic**. If the coefficient of restitution is 0, then the collision is **completely inelastic**.

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$$e = -\frac{v_{2A} - v_{2B}}{v_{1A} - v_{1B}}$$

v_{2A} – final velocity of object 1
 v_{2B} – final velocity of object 2
 v_{1A} – initial velocity of object 1
 v_{1B} – initial velocity of object 2

Another way to get the coefficient of restitution is in terms of the initial height and the bounce height:

$$e = \sqrt{\frac{h_{\text{bounce}}}{h_{\text{initial}}}}$$

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